CERb – Liquid Diffusion Coefficient Apparatus

The liquid diffusivities coefficient apparatus has been designed to enable measurement of molecular diffuivities relating to an equimolar counter-diffusion

PRODUCT CODE: **CERB**

Categories: CE Series - Chemical Engineering, CERb - Liquid Diffusion Coefficient Apparatus, Chemical, Educational

Tags: Bench Mounted , CE Series , CERb , Chemical Engineering Basic Process Principles , Equimolar Counter-Diffusion , Fick's Law , Liquid Diffusion Coefficient Apparatus , Magnetic Stirrer , Mass Transfer Theory , Molecular Diffuivities

DESCRIPTION

The liquid diffusivities coefficient apparatus laboratory equipment has been designed to enable measurement of molecular diffuivities relating to an equimolar counter-diffusion process and in so doing, familiarise students with the basic notions of mass transfer theory.

Armfield has developed a unique diffusion cell which overcomes the traditional problem of slow diffusion rates in liquids requiring long observation times but without sacrificing accuracy or introducing convective effects. Essentially, the cell consists of a honeycomb of accurately dimensioned capillaries positioned between two liquids of differing concentration of the solute, whose diffusion coefficient is to be determined.

In practice, a small volume of concentrated solution is placed on one side of the honeycomb, whilst the other side consists initially of a large volume of pure solvent (water). As diffusion of the solute occurs, the concentration within the larger volume increases and is monitored with a conductivity sensor and meter.

The mixture is continuously stirred with a magnetic stirrer to ensure uniform concentration within the bulk liquid.

Whilst the conductivity sensor may be readily calibrated for any required aqueous system, for introductory studies dilute solutions of sodium chloride are recommended, for which conductivity data is provided.

TECHNICAL SPECIFICATIONS

Diffuser vessel: Capacity 1l
Conductivity meter: 3 ranges 199.9µ5 to 19.99mS
Computer output: RS232

FEATURES & BENEFITS

Ordering specification

- Bench mounted apparatus for the determination of diffusion coefficients of components in the liquid phase. The method employs a diffusion cell of capillary tubes so constructed to permit equimolar counter diffusion between liquids of differing concentration each side of the cell without convective effects being present
- Concentration changes on one side of the cell with respect to time are measured with the conductivity cell and the meter provided and a magnetic stirrer keeps the bulk solution well mixed
- Possible to obtain reproducible and accurate values of diffusivity within a period of 1.5 hours of practical laboratory time
- Battery powered

Recommended accessories

Stop clock

Cartridge dioniser

Requirements

Electrical supply:

Battery operated

Software requires the user to have a PC running Windows 7 or above with a USB port.

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

Volume: 0.10m³

Gross Weight: 10kg

Dimensions

Overall dimensions

Height: 0.31m

Diameter: 0.19m

Conductivity meter

Length: 0.25m

Width: 0.20m

Height: 0.13m

Order Codes

CERb

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